

Universal Electronic Visit Verification (EVV) Platform

A generalized EVV (Electronic Visit Verification) platform can evolve beyond healthcare into a full-scale enterprise workforce verification and field operations ecosystem. This document outlines the architecture, modules, features, industries, compliance structure, technology stack, and strategic positioning of a universal EVV platform.

Industries That Can Use the Platform

- Healthcare & Caregiving
- Security Industry
- Cleaning & Facility Management
- Construction & Field Workers
- Delivery & Logistics
- Education & Community Services
- Government & Municipal Services
- Telecom & Technical Support
- Sales & Merchandising

Core Platform Layers

- Identity & Authentication Layer
- Multi-role authentication
- Biometric login
- RBAC (Role-Based Access Control)
- Enterprise isolation

Multi-Tenant SaaS Architecture

- Shared DB + Tenant ID
- Separate Schema
- Separate Database
- Recommended: Separate schema initially with enterprise DB upgrades

Workforce Management Engine

- Employee onboarding
- Contractor management
- Skill certifications
- Shift scheduling

- Attendance
- Geo-fencing

EVV Verification Engine

- GPS verification
- Biometric verification
- NFC / RFID
- QR code verification
- Telephony verification
- Beacon verification
- Device fingerprinting

Scheduling & Dispatch System

- Smart scheduling
- Route optimization
- Recurring visits
- Emergency dispatch
- AI workforce allocation

Mobile Workforce Application

- Clock in/out
- Offline sync
- Incident reports
- Media uploads
- Signature capture
- Panic/SOS button

Supervisor Dashboard

- Live worker map
- Compliance alerts
- Productivity metrics
- Payroll estimation

Compliance & Audit Engine

- HIPAA
- POPIA
- GDPR
- SOC2

- ISO 27001
- Fraud detection

AI & Intelligence Layer

- Predictive scheduling
- Fraud detection AI
- Workforce optimization
- Compliance risk scoring
- AI-generated reports

Client & Customer Portal

- Track visits
- Approve services
- Digital signatures
- Ratings & complaints

Payroll & Billing Engine

- Automated timesheets
- Overtime calculations
- Invoice generation
- Subscription billing

Reporting & Business Intelligence

- Attendance reports
- Revenue analytics
- SLA compliance
- GPS route history
- Excel/PDF export

Communication System

- Push notifications
- SMS
- Email
- In-app chat
- Emergency alerts

API & Integration Layer

- Workday
- SAP SuccessFactors
- Sage
- QuickBooks
- Google Maps Platform
- Mapbox
- HL7/FHIR integrations

Enterprise Security Architecture

- AES-256 encryption
- TLS 1.3
- Zero trust architecture
- WAF
- DDoS protection
- SIEM integration

Recommended Technical Stack

- Frontend: React, Next.js, React Native
- Backend: Node.js, NestJS, Go
- Database: PostgreSQL, Redis, TimescaleDB
- Infrastructure: Kubernetes, Docker, Terraform
- AI: Python, TensorFlow, OpenAI APIs

Modular Enterprise Design

- Healthcare EVV
- Guard Patrol
- Field Technician
- Delivery Tracking
- Cleaning Verification

Revenue Model

- SaaS subscription
- Per-user pricing
- Enterprise licensing
- Usage-based billing
- White-label licensing

Future Expansion Possibilities

- AI agents
- Smart wearables
- Drone verification
- Blockchain audit logs
- Digital identity wallets

Some Known EVV Software Providers

Provider	Website
Sandata Technologies	https://www.sandata.com
HHAeXchange	https://www.hhaexchange.com
CareSmartz360	https://www.caresmartz360.com
AlayaCare	https://www.alayacare.com

Final Strategic Insight

A generalized EVV platform can become:

“The operating system for field workforce accountability.”

Healthcare EVV is only the entry point. The broader long-term opportunity is creating a unified enterprise platform for workforce verification, compliance, scheduling, and intelligent field operations across multiple industries worldwide.